

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

Salesforce Flow Automation • Functional, Performance & UAT Documentation

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Note: Page numbers can be refreshed after final editing in Word.

1. Project Overview

The project streamlines the procurement of standard laptops by replacing repetitive manual coordination with a Salesforce Flow. Employees submit a laptop request using a standardized request form. The Flow validates the request, determines whether the request meets the standard configuration, routes approval when required, creates the procurement record, and sends notifications to the relevant stakeholders.

The solution is designed for standard laptop orders where the organization has predefined device models, approved accessories, cost limits, and approval rules. Non-standard or incomplete requests are routed to an exception path instead of being automatically ordered.

Project Information

Parameter	Value
Automation Platform	Salesforce Flow
Primary Process	Standard Laptop Procurement
Primary Users	Employees, Managers, IT Procurement, IT Support
Trigger	New or submitted laptop request
Primary Output	Validated procurement request / order record
Notification	Email / Salesforce notification

2. Problem Statement

- Laptop procurement can involve repeated data entry, manual validation, approval follow-ups, and status communication.
- Manual processing increases the risk of incorrect device selection, missing information, duplicate requests, delayed approvals, and inconsistent records.
- The proposed Flow centralizes the business rules and automates the routine steps so that standard laptop requests can be processed consistently and traceably.

3. Objectives

Objective	Description
Standardize requests	Capture employee, department, location, device type, and justification in a consistent format.
Automate validation	Check mandatory fields, approved models, budget thresholds, and duplicate requests.
Automate approvals	Route requests to the appropriate manager or procurement approver based on defined rules.

Create records automatically	Create or update procurement/order records after successful validation and approval.
Improve visibility	Maintain status, timestamps, owner, and audit information in Salesforce.
Reduce manual effort	Use Flow to handle routine routing, notifications, and record updates.

4. Scope

In Scope

- Standard laptop request intake and validation.
- Approved laptop model and configuration validation.
- Budget / cost threshold checks.
- Duplicate request check.
- Approval routing and status updates.
- Procurement record creation and notifications.
- Exception handling for incomplete or non-standard requests.
- Functional, performance and user acceptance testing.

Out of Scope

- Direct vendor payment processing.
- Warehouse hardware installation and imaging.
- Non-standard custom laptop engineering.
- Physical delivery tracking unless a separate integration is configured.

5. Proposed Solution & Flow Design

A record-triggered Salesforce Flow is recommended for the automation. The Flow starts when a laptop request is submitted or reaches the status configured for processing.

End-to-End Process

Step	Flow Stage	Description
1	Employee submits request	Required employee, department, location, laptop model and business justification are captured.
2	Validate request	Flow checks mandatory fields, approved model, budget and duplicate request conditions.
3	Determine approval	Flow evaluates whether manager / procurement approval is required.

4	Approval	Request is routed to the appropriate approver and status is updated.
5	Create procurement record	After approval, Flow creates or updates the procurement order/request record.
6	Notify stakeholders	Employee, manager and procurement team receive status notifications.
7	Close / hand off	Request is marked Ready for Procurement or Completed based on the configured process.

Flow Logic – High-Level Design

- Start → Laptop Request created/submitted
- Get Request details → Employee / Department / Model / Cost / Status
- Decision: Are mandatory fields complete?
- No → Set status = Needs Information → Notify requester → End
- Yes → Decision: Is laptop model approved?
- No → Set status = Exception Review → Notify procurement → End
- Yes → Decision: Is duplicate active request found?
- Yes → Set status = Duplicate / Review → Notify requester → End
- No → Decision: Does cost exceed approval threshold?
- Yes → Submit for manager/procurement approval
- No → Auto-approve standard request
- Create/update Procurement Order → Notify stakeholders → End

6. Business Rules and Automation Logic

Rule ID	Rule	Implementation
BR-01	Mandatory Fields	Employee, department, location, laptop model, quantity and business justification must be present.
BR-02	Approved Model	Only laptop models marked as Active/Approved can follow the standard automated path.
BR-03	Quantity	Quantity must be a positive whole number and must not exceed the configured request limit.
BR-04	Duplicate Check	An active request for the same employee and device category should be flagged for review.
BR-05	Approval Threshold	Requests above the configured cost threshold require approval.

BR-06	Non-Standard Request	Requests outside the approved catalog are routed to Exception Review.
BR-07	Notification	Status changes generate notifications to the relevant requester, approver or procurement user.
BR-08	Auditability	The Flow records status, owner and relevant timestamps for traceability.

7. Data Model and Configuration

The following logical objects/fields can be implemented using Salesforce custom objects, standard objects, or an existing procurement data model.

Record / Object	Important Fields
Laptop Request	Request ID, Employee, Department, Location, Model, Quantity, Justification, Cost, Status
Laptop Catalog	Model, Manufacturer, Specification, Unit Cost, Active, Standard/Non-Standard
Approval	Request, Approver, Approval Level, Decision, Decision Date, Comments
Procurement Order	Order ID, Request, Vendor, Quantity, Total Cost, Order Status, Order Date
Notification Log	Request, Recipient, Notification Type, Sent Date, Delivery Status

Recommended Flow Elements

- Start / Record Trigger
- Get Records – retrieve employee, catalog and existing request data
- Assignment – prepare calculated values
- Decision – validate request and select processing path
- Create Records – create procurement request/order
- Update Records – update request and approval status
- Approval / Action – route approval where applicable
- Email Alert / Notification – communicate status
- Fault Path – capture errors and notify support

8. Functional Test Scenarios & Results

The following test cases are suitable for the functional testing template shown in the reference material. Actual results and screenshots should be updated after execution.

Test Case ID	Scenario	Test Steps	Expected Result	Sample Result	Screenshot
FT-01	Valid standard laptop request	Enter all valid required values and submit.	Request accepted; standard Flow path starts.	Pass	
FT-02	Missing mandatory field	Leave laptop model or employee field blank and submit.	Request is rejected or marked Needs Information; user receives message.	Pass	
FT-03	Invalid / non-standard model	Select a model not marked Approved.	Request is routed to Exception Review.	Pass	
FT-04	Duplicate request	Submit a second active request for the same employee/device category.	Duplicate is detected and request is flagged for review.	Pass	
FT-05	Approval required	Submit request above the configured cost threshold.	Approval request is routed to the correct approver.	Pass	
FT-06	Approval granted	Approver approves the request.	Status changes to Approved and procurement record is created/updated.	Pass	
FT-07	Approval rejected	Approver rejects the request.	Status changes to Rejected; requester is notified.	Pass	
FT-08	Notification	Trigger key status changes.	Correct stakeholders receive the expected notification.	Pass	
FT-09	Quantity validation	Enter zero, negative or excessive quantity.	Validation prevents invalid quantity and displays a clear message.	Pass	
FT-10	Flow fault handling	Simulate record creation/update failure.	Fault path captures error and alerts support; request is not silently lost.	Pass	

9. Performance Test

Performance testing should verify that the automation completes within an acceptable time and does not create excessive Salesforce processing, query, or notification overhead. The exact thresholds should be confirmed against the project's Salesforce environment and data volume.

Test ID	Scenario	Test Method	Expected Result
PT-01	Single request processing	Submit one valid standard request.	Flow completes promptly and reaches expected status.
PT-02	Concurrent requests	Submit multiple standard requests within a short period.	Requests are processed without failures or unexpected delays.

PT-03	Large catalog lookup	Process a request against a realistic catalog size.	Catalog lookup remains responsive and returns the correct model.
PT-04	Duplicate check	Run request with many historical records.	Duplicate check remains accurate without unnecessary record retrieval.
PT-05	Notification processing	Trigger multiple status notifications.	Notifications are generated once per required event.
PT-06	Fault recovery	Force a downstream record/update error.	Fault path executes and preserves useful diagnostic information.

Performance evidence to capture:

- Flow debug run duration / execution details
- Error count and failed transactions
- Record/query behavior for realistic data volume
- Notification delivery evidence

10. User Acceptance Testing (UAT)

UAT confirms that the automated laptop procurement process meets the business requirements from the perspective of employees, managers, procurement users and IT support.

UAT ID	Test Scenario	Acceptance Criteria	Sample Result	Feedback
UAT-01	Employee submits standard laptop request	Request can be submitted with clear guidance.	Pass	
UAT-02	Employee submits incomplete request	System identifies missing information.	Pass	
UAT-03	Manager approves request	Approval is easy to locate and status updates correctly.	Pass	
UAT-04	Manager rejects request	Rejection is recorded and requester is informed.	Pass	
UAT-05	Procurement reviews approved request	Procurement user can identify all required order details.	Pass	
UAT-06	Exception request	Non-standard request is clearly routed for manual review.	Pass	
UAT-07	Request status tracking	Stakeholders can see the current request status.	Pass	
UAT-08	Audit information	Relevant dates, owner and approval information are retained.	Pass	

UAT Sign-Off

Tester Name	
Business Owner	
Date	
Signature	

11. Defect Tracking and Acceptance Report

Defect Tracking Template

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Example: duplicate request not detected	Submit two active requests for same employee/model.	Medium	Open	
BG-002	Example: notification not delivered	Trigger approval decision and verify recipient.	Low	Open	

Acceptance Summary

Section	Total	Passed	Failed	Not Tested
Functional test cases	10	10	0	0
UAT scenarios	8	8	0	0
Performance scenarios	6	6	0	0

Sample acceptance statement: The standard laptop procurement automation is considered ready for controlled deployment when all critical functional and UAT cases pass, no unresolved high-severity defects remain, and the business owner approves the final process.

12. Security, Controls and Exception Handling

Control	Implementation
Access Control	Only authorized users should create, approve, or modify procurement requests.
Least Privilege	Flow actions should operate with only the access required for the business process.
Data Validation	Use validation rules and Flow decisions to prevent incomplete or invalid requests.

Audit Trail	Maintain request status, owner, approval decision and relevant timestamps.
Fault Handling	Use Flow fault paths to capture errors and notify support rather than silently failing.
Duplicate Prevention	Check for existing active requests before creating a new procurement record.
Sensitive Data	Avoid storing unnecessary personal or financial information in the request.

13. Deployment Checklist

- ☐ Confirm approved laptop catalog and pricing are current.
- ☐ Confirm request fields and validation rules are active.
- ☐ Confirm Flow entry criteria and decision rules.
- ☐ Confirm approval routing and approver assignments.
- ☐ Confirm email templates / notifications.
- ☐ Confirm fault paths and error logging.
- ☐ Run functional regression tests in the target environment.
- ☐ Run UAT and obtain business owner sign-off.
- ☐ Document rollback / disable procedure.
- ☐ Monitor initial production transactions after deployment.

14. Conclusion

The proposed Salesforce Flow provides a structured approach to automating standard laptop procurement. By validating requests, routing approvals, creating procurement records, and notifying stakeholders, the process can reduce repetitive manual work while improving consistency, traceability and visibility.

The documentation provides a ready-to-use basis for project development, functional testing, performance testing and UAT. Project-specific screenshots, actual execution times, Salesforce Flow debug evidence, defect results and final sign-off should be added during execution.

Appendix A – Suggested Salesforce Flow Build Sequence

1. Create / confirm Laptop Request fields.
2. Create / confirm Laptop Catalog records and Approved flag.

3. Create record-triggered Flow on Laptop Request.
4. Get related employee and laptop catalog data.
5. Validate required fields and quantity.
6. Check approved model and duplicate active request.
7. Calculate total estimated cost.
8. Evaluate approval threshold.
9. Route approval when required.
10. Create/update Procurement Order after approval.
11. Update request status and timestamps.
12. Send notifications.
13. Add fault paths and error notifications.
14. Activate only after functional and UAT approval.

Appendix B – Screenshot Evidence Checklist

- Flow Builder canvas showing the complete automation.
- Laptop Request record before submission.
- Validation / error message for an invalid request.
- Approval request and approval decision.
- Created Procurement Order record.
- Notification email / Salesforce notification.
- Flow Debug Details showing a successful run.
- Fault-path or exception handling evidence.

END OF DOCUMENT